

Handling of Large File in Fedora

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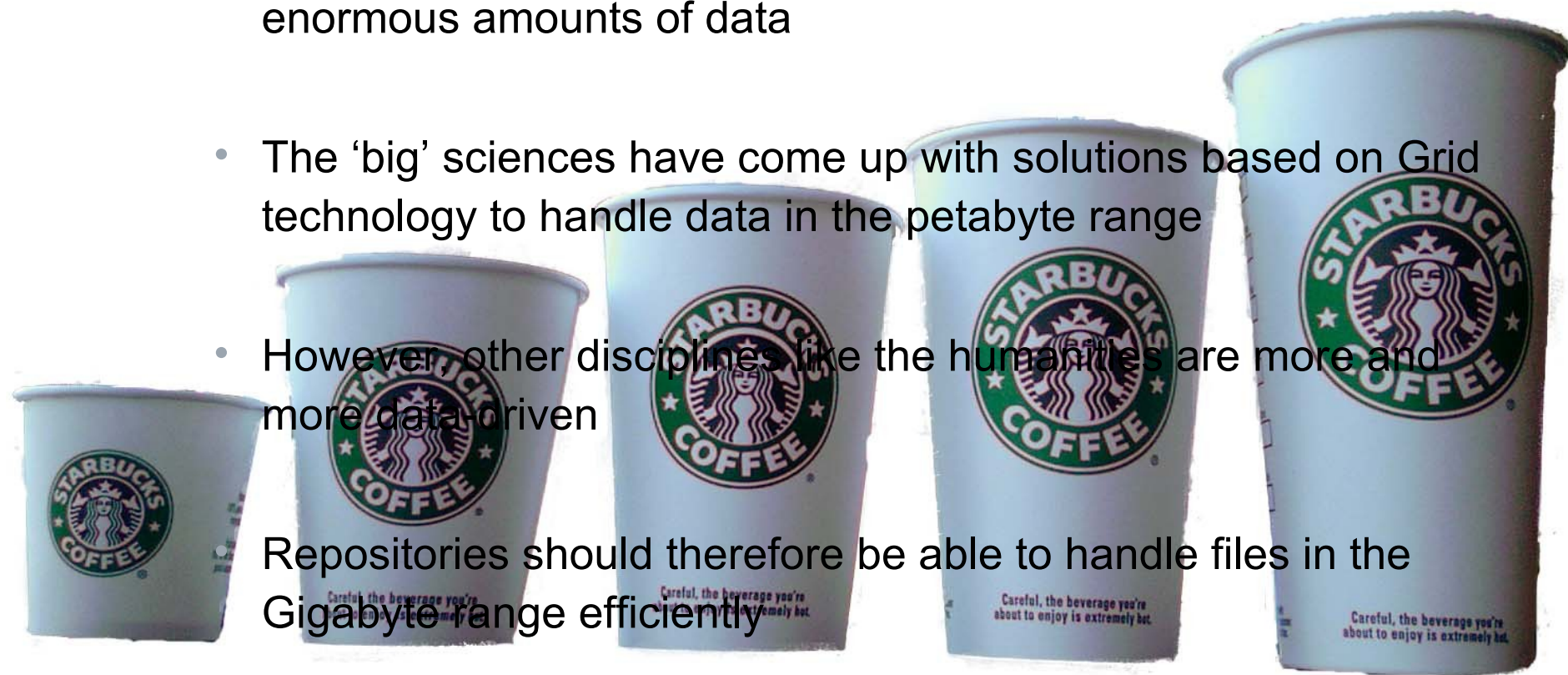
Agenda

- **Motivation**
- Approach
- Results
- Status

Motivation: New Challenges for Repositories

- In an e-Science context, repositories have to be able to handle enormous amounts of data
- The 'big' sciences have come up with solutions based on Grid technology to handle data in the petabyte range
- However, other disciplines like the humanities are more and more data-driven

Repositories should therefore be able to handle files in the Gigabyte range efficiently



Motivation: Use Case from the Humanities

- Max Planck Institute for the History of Science hosts various collections of primary and secondary materials
- Most of the data is high-resolution scans of manuscripts, photographs of artwork and historic artifacts
- File sizes for master scans/reproductions reach 1 Gigabyte
- Additional image resolutions and metadata adds to the total amount of curated data
- This sums up to several 10,000 objects with a total size of ~28 Terabyte

Motivation: Use Case from the Humanities /2

- When working with the primary data, scholars are often only interested in parts of the data (e.g., for annotating a part of an image)
- In many cases, they are interested in manipulated versions (e.g., scaling, cropping)
- These manipulations are dependent on the object type
- Providing scholars a tool that supports them in their daily work therefore requires a efficient ingestion and dissemination architecture

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Approach: Goals

- Adhere to Fedora's principle of datastreams
- Improve read/write performance for large content
- Optimize efficiency by allowing for random file access, thus
 - reducing the processing resources required to provide a dissemination (not stream the whole file if only the last few bytes are needed)
 - saving network bandwidth by generating the dissemination physically next to the storage system

Approach: Current Situation

- Remote datastreams are not managed by Fedora
- Locally managed content:
 - Inline XML
 - Managed Content
- Externally managed content
 - External Referenced Content
 - Redirect

Approach: New Control Group Type

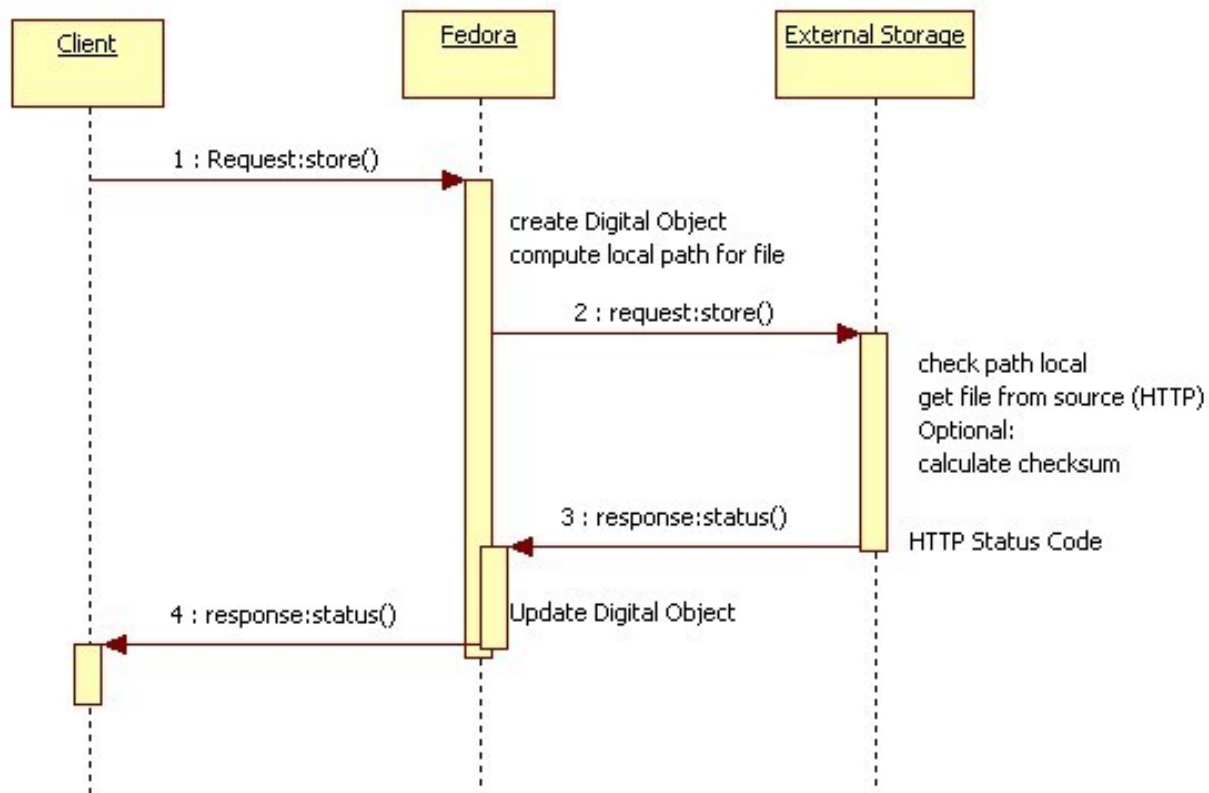
- Introduce a new Control Group Type to manage content on remote systems: **Managed External**
- Content of this Control Group Type are managed (ingest/retrieve) by Fedora - but the files are stored on remote system.
- Fedora mediates access to the remote storage to avoid the duplication of existing functionality, thus circumventing the need for yet another access management component

Approach: Requirements

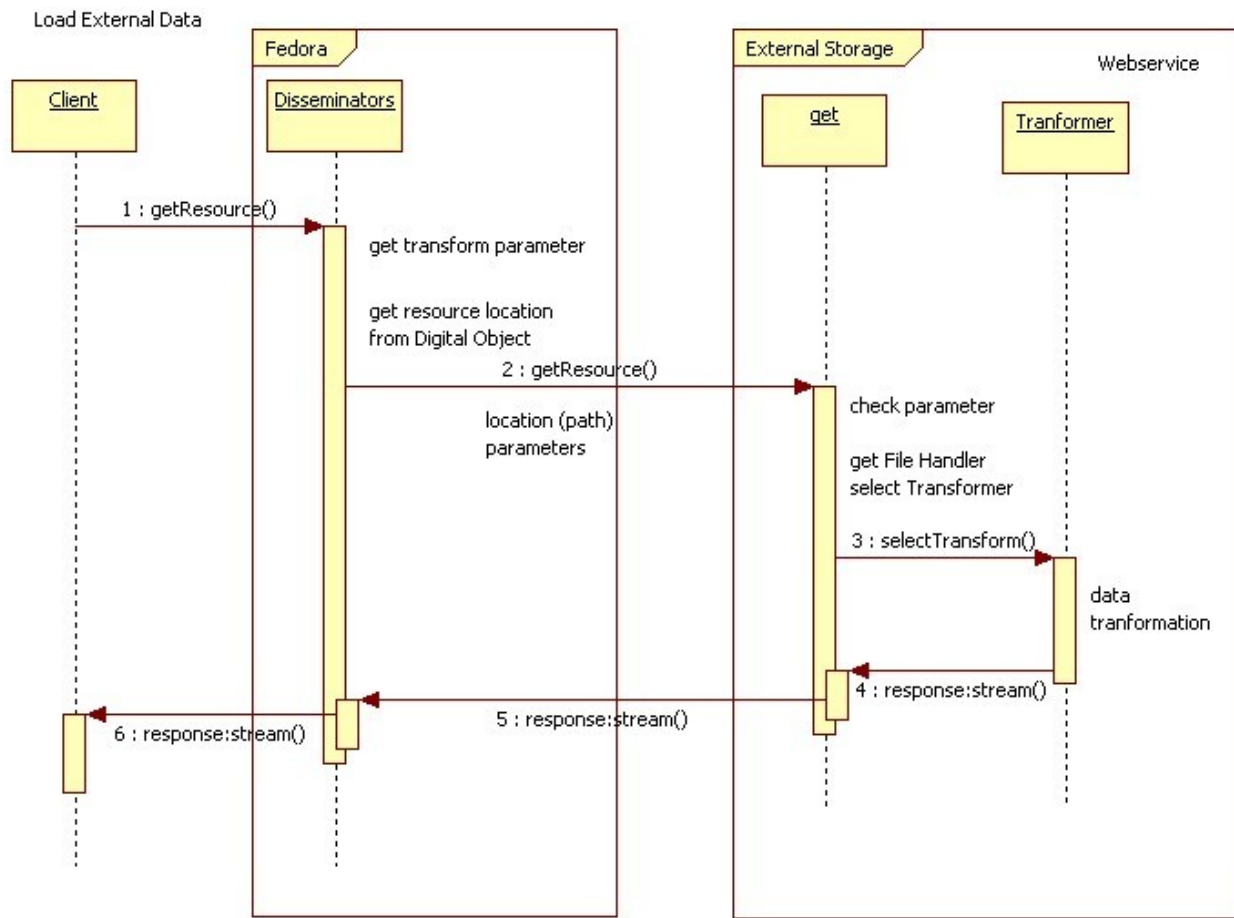
- Implementation as a web service with simple interface
 - put – store stream to file
 - get – open file and deliver as stream
 - delete – delete files (purge)
- Store datastream under a path and filename provided by Fedora (no own data management)
- No version control, instead reuse Fedora's versioning functionality for binary content

Approach: Write Sequence

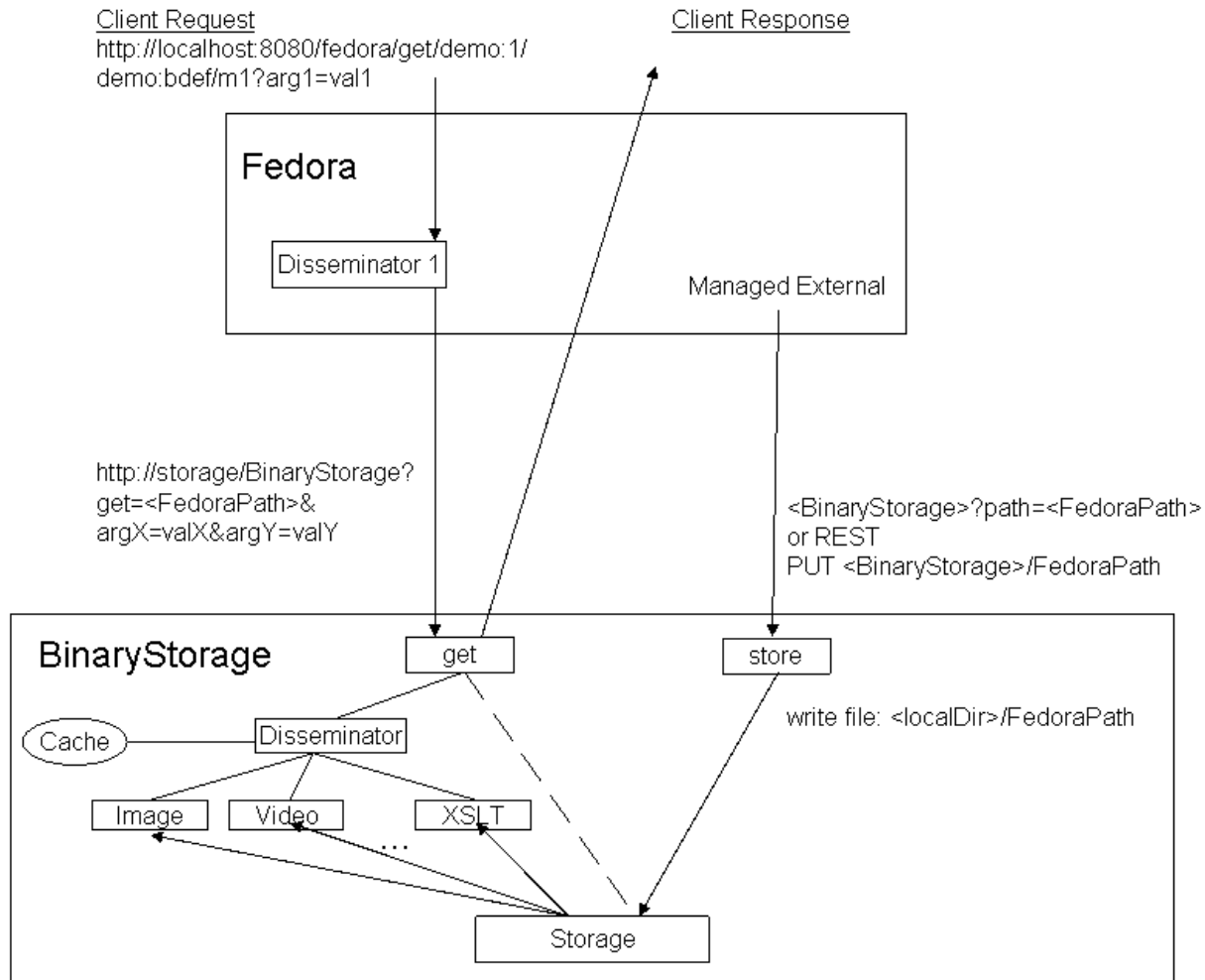
Store Data External



Approach: Read Sequence



Approach: Schematic View



Approach: New Disseminator Type

- Fedora Disseminators operate on HTTP data streams
 - + very flexible
 - no random file access
 - always transfer whole file, even if only parts are requested
- Remote Storage System with new Disseminators
 - + direct random file access
 - + high throughput
 - limited to local files
 - restricted chaining

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Results: Measurements for Write Access

Time to store data in the repository

Size	Fedora (managed)	External	Ratio
1 GB	150 s	87 s	1.7

Results: Measurements for Read Access

Time to stream data to client, without transforms measured

Size	Fedora (managed)	External	Ratio
2 MB	0.86 s	0.14 s	6.0
50 MB	5.38 s	3.04 s	1.7
1 GB	138.76 s	113.96 s	1.2
2 GB	N/A	271.41 s	N/A

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Status: Changes in Fedoras Implementation

- Implementation of the new Control Type is not finished yet
- Fedora source code classes altered so far:
 - DefaultAccess
 - BasicDigitalObject
 - DefaultManagement
 - BaseTripleGenerator
- Internal designator for ‚Managed External‘ is ‚S‘ (Stored External)

Status: Implementation of new Disseminator

- New Parm Type – FILE
- Path to externally stored file is provided to Disseminator (instead of the datastream)
- Developers need to be aware of the differences between Disseminators for datastreams (traditional) and for direct file access (new)
- Restriction for chaining the new Disseminator type: must be the first in a chain

Status: Implementation of External Storage

- Current implementation of proposed external storage: BinaryStorage
- Implemented as Servlet with functions for storing, retrieving and deleting files
- Disseminator for direct file access, e.g. transform images, retrieve video sequence

Control Group in Fedora Admin

Object - changeme:3

Properties Datastreams Disseminators

DC
New...

ID

Control Group

☐ Internal XML Metadata

☐ Managed Content

☐ External Referenced Content

☐ Redirect

☒ Managed External

Fedora will store content to an external system. But the management of content is still part of Fedora. This is useful for large binary content if the external system supports content type specific data manipulation. If you store large images in with fedora and different representation are permanently requested so is a special image storage with local disseminator the better choice. For default this option link to the BinaryStorage. The content is still stream through Fedora but only in the requested representation (by saving bandwidth).

State: Active

Versionable: Updates will create new version

MIME Type: text/xml

Label: Enter a label here.

Format URI:

Alternate IDs:

Checksum: Default

Location Source: http://

Target: calhost:8080/BinaryStorage/servlet/BinaryStorage.ManageData

View

Save Datastream

Thanks!